

Affinity Ultrafiltration for Protein Purification

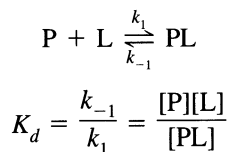
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Purification of a biochemical product from fermentation broths or other complex biological mixtures generally involves a combination of techniques which resolve substances according to differences in their physicochemical properties. These techniques have traditionally been optimized individually rather than as a part of the integrated, continuous process. Moreover, a lot of the development and refinement of the procedures has taken place on a laboratory scale, with little consideration given to the economics of scaling up. As a result, downstream processing presently accounts for a major portion of the processing costs of producing a biotechnological product.

In order to cut down on costs and also to prevent the product loss that occurs during several purification stages, it is critical to design processes that possess high product recovery and resolution from impure feed streams.

1. AFFINITY CHROMATOGRAPHY

Among the available purification strategies, affinity chromatography is known to be the most powerful in terms of specificity. This has been amply demonstrated in the present volume. Affinity chromatography makes use of the natural affinity displayed between a biological macromolecule and a complementary ligand, for instance, a substrate analog or an inhibitor:



where K_d is the dissociation constant; k_1 and k_{-1} are forward and reverse rate constants, respectively; and $[P]$, $[L]$, and $[PL]$ are the respective concentrations of the protein, the

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ligand, and the affinity complex. An extremely stable complex is represented by a low K_d value, and vice versa.

In affinity chromatography, the ligand is immobilized onto a solid matrix. After the protein–ligand affinity complex is formed and the impurities have been washed away, the bound protein molecule is dissociated from the ligand. This is achieved by a change in pH and/or ionic strength, by using chaotropic agents, or, in a more specific manner by using a solution of a free complementary ligand.

Despite the potential of this technique, its scaled-up application has been limited because of the high material costs, the requirement for sample pretreatment, and other problems characteristic of chromatographic separations. Currently, it is used only as a final step in the downstream processing of high-value products.

There have been attempts during the last decade to exploit affinity interactions in a mode which could be used early during the downstream processing of a protein. Such a configuration should be amenable to easy scale-up while at the same time overcoming the limitations of the chromatographic mode of operation.

2. ULTRAFILTRATION

Ultrafiltration is based on the ability of a membrane to separate molecules from other molecules or particles on the basis of size. Membrane filtration is an established technology in biochemical and pharmaceutical industries. In biotechnology, it has been integrated into downstream processing at various stages such as cell harvesting, separation of biomolecules, and concentration and diafiltration of products. It is simple, rapid, and economical and has low energy requirements.

Developments in polymer chemistry and module construction have resulted in the availability of an array of ultrafiltration membranes of high chemical and physical stability combined with high fluxes, for different scales of operation. Filtration is most effective when operated as a tangential flow (also termed cross-flow filtration) such that the flow of the feed is parallel to the membrane surface and is recirculated rapidly enough to prevent concentration polarization.

As a separation technique, ultrafiltration is limited by its low resolving power. At least a tenfold difference in molecular weight is necessary for separation of molecules.

3. AFFINITY ULTRAFILTRATION

In order to exploit the high-volume processing capability of membrane filtration, and at the same time make it extremely selective, the concept of affinity ultrafiltration has been developed. This technique works on the principle that when a protein to be purified is free in solution, it passes unimpeded through the pores of the ultrafiltration membrane, whereas when it is bound to a macromolecular affinity ligand placed on one side of the membrane, it is restricted to that side of the membrane (Mattiasson and Ramstorp, 1983; Mattiasson *et al.*, 1983; Mattiasson and Ling, 1986; Luong *et al.*, 1987). Compounds which do not bind the ligand pass freely through the membrane and are thus separated from the protein–ligand complex. After all the undesired components have been removed, the protein is desorbed from the ligand by the addition of an appropriate dissociating medium

and is then recovered in the filtrate. The ligand is regenerated by removing the dissociating medium and recycled. The general procedure of affinity ultrafiltration is schematically represented in Fig. 1. This process can be run as a batch, semibatch, or continuous operation.

The respective sizes of the target molecule and the macromolecular ligand together with the cutoff of the filtration membrane determine the efficiency and quality of purification. The low selectivity of ultrafiltration membranes makes it imperative that the molecular weight of the macroligand be more than ten times that of the protein and the contaminant species.

The pores of the membrane should be of a size and uniformity which prevent the leakage of the ligand and, at the same time, should be highly permeable to the free protein and the impurities under the operating conditions. Therefore, prior to the affinity step, it is advantageous to study the behavior of the macromolecular ligand and the target protein individually on ultrafiltration, through the chosen membrane (Luong *et al.*, 1988a).

Adsorption of the macromolecular ligand to the membrane may lead to reduced filtration rates, and it may deposit as a layer on the membrane surface with its own rejection properties. It may be possible to minimize these interactions by introducing charges on the membrane surface. Protein-protein interactions such as complexation may also lead to enhanced rejection coefficients during protein ultrafiltration. The significance of the latter phenomena were well demonstrated in a study on the isolation of urokinase by affinity

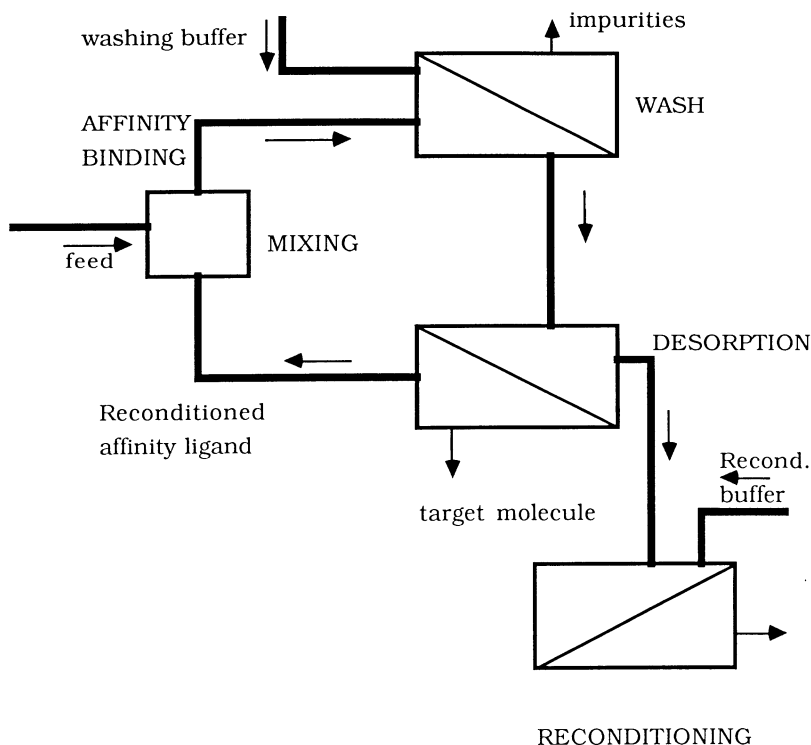


FIGURE 1. Schematic of the experimental setup for membrane affinity filtration.

filtration (Male *et al.*, 1990). When the enzyme, with a molecular weight of 31,000, was ultrafiltered in the absence of the affinity polymer using 10,000- or 100,000-MW-cutoff Millipore systems, it was not detected in either the filtrate or the retentate, suggesting that it was somehow binding to the polysulfone membrane. However, the enzyme was able to pass through a 100,000-MW-cutoff Amicon YM membrane (cellulose acetate) with greater than 95% recovery and was retained to a level of more than 95% by a 10,000-MW-cutoff Amicon membrane. Furthermore, during the purification of urokinase from human urine, it was noticed that the high-molecular-weight materials present in this crude source apparently formed a complex with the enzyme and prevented its passage through a 100,000-MW-cutoff membrane. The complex was dissociated by the addition of salt (2M NaCl), resulting in 70% urokinase activity in the filtrate.

3.1. Affinity Binding

Binding between the ligand and the material to be purified can be achieved in one of the following ways:

1. by mixing together the crude solution with the ligand in the ultrafiltration unit or in a separate mixing chamber; or
2. by pumping the crude solution and the solution containing the ligand on either side of the filtration membrane and letting the binding take place when the protein passes over to the side where the ligand is located.

The first alternative is the method of choice for relatively clarified broths or when the solutions containing soluble and insoluble high-molecular-weight contaminants have been preultrafiltered by using a membrane with a suitable molecular weight cutoff. Most of the studies reported on affinity ultrafiltration have employed this procedure.

The second method is a more general approach and could be directly integrated with, for example, a fermentation broth or other complex mixtures of biological molecules. Here, the membrane forms a protective barrier for the ligand against particulate matter, infectious agents, and so on.

The purification of *Bacillus licheniformis* α -amylase by the latter approach with the use of a cross-linked starch slurry as the affinity adsorbent was studied by Somers and Van't Riet (1990). Recovery on the basis of enzyme activity was reported to be 95%.

The binding process during affinity ultrafiltration is influenced to a great extent by the concentration of the macromolecular ligand. The binding capacity initially increases with increasing ligand concentration but reaches a plateau at high concentrations as ligands become sterically inaccessible for protein binding. A similar phenomenon has been observed during affinity chromatography.

3.2. Washing

Washing of the affinity complex is usually performed in the diafiltration mode such that the volume of the retentate is maintained constant by adding the washing buffer at a rate equal to the filtration rate.

When affinity interactions are used in a chromatographic purification process, the binding strength only needs to be sufficient to provide effective separation so that the eventual dissociation of the complex to obtain the pure product is not difficult. However, in affinity ultrafiltration, washing is far more intense so that anything that is not firmly bound

is washed away. The washing step begins with an equilibrium mixture of free and bound protein. As the washing proceeds, the free protein concentration decreases and the bound protein releases from the adsorbent in an effort to reestablish equilibrium. One strategy could be to use a high filtration rate to minimize loss of the bound component during the washing step. However, for very low affinity systems, this would result in a significant loss of the desired protein. According to Ling and Mattiasson (1989), if the binding between ligand and target protein is assumed to be in equilibrium, to get a calculated recovery of more than 90% when at least 99% of other material is washed out, the effective binding strength must be higher than $10^6 M^{-1}$. However, experimental and modeling work presented by Herak and Merrill (1989) using Cibacron blue-agarose and human serum albumin (HSA) indicated that the release of the target protein during the washing step is better represented by a model where no HSA is released from the gel. The observed slow kinetics of the release of the adsorbed protein during the washing step could be applicable to most other systems and may be exploited to minimize loss of the target protein.

3.3. Elution

For the elution of the bound protein, the buffer is changed so that the binding interaction is unfavorable and the complexed protein is released. Affinity between the protein and the ligand will change gradually as the elution buffer replaces the coupling buffer. The important aspect of this step is that the bound protein is readily released without an excessively large volume of buffer. Moreover, the conditions used must be favorable for the biomolecule, as well as the membrane. As in affinity chromatography, elution can be performed either by use of a competitive ligand or by altering pH, ionic strength, etc. The first alternative is attractive, since it is performed under conditions favorable for binding and preserves the biological activity of the molecule and also does not negatively influence the nature of the membrane. The drawbacks might be the high cost involved in using a competitive affinity ligand and the fact that the product is contaminated with the free ligand. Nonspecific elution, on the other hand, is cheap but may often be harmful either to the biomolecule to be purified or to the membrane. However, the choice of elution strategy will very much depend on the system concerned.

3.4. Reconditioning

After the elution step, the free macromolecular affinity ligand is reconditioned generally in a separate membrane filtration step where the dissociating buffer is replaced with the fresh binding buffer by diafiltration. It is then ready to be used for purification of subsequent batches.

The eluate containing the purified product is very dilute. Passage through another ultrafiltration membrane concentrates the product and at the same time removes the free competing ligands on the basis of molecular size.

4. MACROMOLECULAR AFFINITY LIGANDS USED FOR AFFINITY ULTRAFILTRATION

The choice of macromolecular affinity ligand could be considered to be the most important parameter determining the applicability of an affinity ultrafiltration system. It

is important that the macroligands are either soluble or easily suspended in aqueous solutions so that they can be pumped through the filtration apparatus. The desirable characteristics of a macromolecular ligand are that it is relatively inexpensive, possesses a large specific surface area, and exhibits negligible nonspecific adsorption of undesired contaminants. Moreover, it should be stable under the required adsorption and elution conditions and show no attrition under high shear forces.

By and large, finding an affinity ligand possessing the molecular size desired in operating affinity ultrafiltration is difficult. This limitation is overcome by "tailoring" the macroligand; that is, a ligand molecule is coupled either to a water-soluble polymer or a water-insoluble carrier that does not biospecifically bind biomolecules. Table 1 lists various macroligands that have been employed for affinity ultrafiltration studies.

4.1. Water-Insoluble Macromolecular Affinity Ligands

One of the earlier studies on affinity ultrafiltration employed heat-killed cells of *Saccharomyces cerevisiae* (mean diameter, 5 μm) as a macroligand for the purification of concanavalin A (Con A; molecular weight, 102,000) from jack bean extract (Mattiasson and Ramstorp, 1984). Con A interacts with the carbohydrate residues exposed on the surface of the yeast cells. The solution containing the cells and the crude Con A extract was pumped through a hollow fiber unit (molecular weight cutoff; 1,000,000) at a flow rate of 500 ml/min. Glucose (150 g/liter) was used to dissociate the affinity complex, and the product stream from the dissociation step was passed through a membrane unit with a molecular weight cutoff of 35,000. The content of glucose was effectively reduced by a diafiltration of the solution. The yeast cell suspension was washed in the reconditioning step so that glucose was eliminated. Con A in a highly purified form was obtained by this procedure at an overall yield of 70%.

As seen in Table 1, carrier particles that are used in affinity chromatography, such as agarose and starch, could also be applicable for affinity ultrafiltration. Particles have been found to be relatively easy to pump through the cross-flow filtration apparatus with minimal fouling of the membranes (Herak and Merrill, 1989). However, particle degradation can be observed with time, depending on the durability of the particles. For example, Sepharose CL-6B, which has a more rigid gel structure consisting of a high cross-link density, was found to be fractured to a lesser extent (5%) after 24 h of circulation than Affi-Gel Blue, which was as much as 30–40% damaged in just 3 h (Herak and Merrill, 1989).

With the use of the commercially available affinity gel *p*-aminobenzyl-1-thio- β -D-galactopyranoside (PABTG) agarose, β -galactosidase was purified from *E. coli* homogenate by the continuous affinity recycle extraction (CARE) process (Pungor *et al.*, 1987).

From experience in classical affinity chromatography, it is known that the available surface area of the particles sets a limit on the amount of ligand and hence the amount of macromolecule that can be bound. The same principle applies to ultrafiltration affinity purification. To have a high binding capacity, the carrier particles must have a large accessible surface area, which means that they should be as small as possible. In this direction, Ling and Mattiasson (1989) attempted to use particles of fumed silica having a diameter of 12 nm and a surface area of 200 m^2/g with immobilized Cibacron blue for affinity ultrafiltration of alcohol dehydrogenase (ADH) and lactate dehydrogenase (LDH) from yeast and porcine heart, respectively. The ligand density was found to be 34 $\mu\text{mol}/\text{g}$ (equivalent to 0.2 $\mu\text{mol}/\text{m}^2$), which was comparable to the density of 0.33 $\mu\text{mol}/\text{m}^2$ for

TABLE 1
Examples of Macromolecular Ligands Used for Protein Purification by Affinity Ultrafiltration

Carrier	Size	Ligand ^a	Target protein ^b	Reference(s)
<i>Insoluble particles</i>				
<i>Saccharomyces cerevisiae</i> cells	5 µm	Surface carbohydrate groups	Concanavalin A	Mattiasson and Ramstorp, 1984
<i>Streptococcus</i> cell groups	1 µm	Surface protein G	Bovine IgG	Chen, 1990
Starch granules	1–10 µm	Cibacron blue	ADH	Mattiasson and Ling, 1986
Silica nanoparticles	12 nm	Cibacron blue	ADH, LDH	Ling and Mattiasson, 1989
Agarose beads	45–165 µm	Cibacron blue	HSA	Herak and Merrill, 1989
Agarose beads	Not stated	PABTG	β-Galactosidase	Pungor <i>et al.</i> , 1987
Sephrose beads	45–165 µm	Heparin	Lactoferrin	Chen, 1990
Sephrose beads	45–165 µm	Protein G	Bovine IgG	Chen, 1990
Liposomes	270 ± 5 Å	Biotin	Avidin	Powers <i>et al.</i> , 1989a,b
Liposomes	600 Å	<i>p</i> -Aminobenzamidine	Trypsin	Powers <i>et al.</i> , 1990
<i>Soluble polymers</i>				
Dextran	MW 2,000,000	Protein A	IgG	Nylén, 1980
Dextran	MW 2,000,000	Estradiol	Δ ⁵⁻⁴ 3-Oxosteroid isomerase	Hubert and Dellacherie, 1980
Dextran	MW 2,000,000	<i>p</i> -Aminobenzamidine	Trypsin	Adamski-Medda <i>et al.</i> , 1981
Dextran	MW 2,000,000	STI	Trypsin	Choe <i>et al.</i> , 1986
Polyacrylamide	MW > 100,000	<i>m</i> -Aminobenzamidine	Trypsin	Luong <i>et al.</i> , 1988a,b; Male <i>et al.</i> , 1987
Polyacrylamide	MW > 100,000	<i>m</i> -Aminobenzamidine	Urokinase	Male <i>et al.</i> , 1990

^aAbbreviations: PABTG, *p*-aminobenzyl-1-thio-β-D-galactopyranoside; STI, soybean trypsin inhibitor.

^bAbbreviations: IgG, immunoglobulin G; ADH, alcohol dehydrogenase; LDH, lactate dehydrogenase; HSA, human serum albumin.

an NAD derivative coupled to silica beads used in high-performance liquid chromatography (HPLC) columns.

The silica nanoparticles formed a colloidal suspension and gave binding kinetics similar to those obtained for binding in free solution. This implied that binding and dissociation occurred almost instantaneously and that the process could be run continuously without the risk of incomplete binding. According to the authors, modification of the silica particles with methoxy-polyethylene glycol (MPEG) to reduce the nonspecific adsorption caused a general decrease in binding capacity to an extent depending on the decrease in the number of available ligands.

The binding of the dehydrogenases with the macroligand was attempted after micro-filtration of the respective homogenates to remove the particulate matter. The recoveries of both the enzymes on a preparative scale in a membrane unit were significantly lower than those obtained on an analytical scale in a test tube. This was attributed to the sensitivity of the enzymes to prolonged filtration. Losses due to adsorption of the enzymes to the filtration membrane in the elution step could also not be ruled out. A significant loss of LDH activity was observed during the washing step, which could be attributed to the low association constant of LDH and Cibacron blue.

The use of affinity-modified liposomes as macroligands for affinity cross-flow filtration systems seems to be quite an interesting approach (Powers *et al.*, 1989a,b, 1990). The unilamellar vesicles (average diameter of 200–700 Å) possess an extremely large surface area; for example, 300 Å-radius liposomes have a specific surface area of 100 m²/ml of liposomes, which, in terms of area per volume of carrier, is approximately the same value as can be obtained with porous silica. The hydrophilic surface of the liposomes results in minimal nonspecific adsorption of undesired impurities, and, because the liposomes remain suspended in solution, membrane polarization and plugging during filtration steps are also avoided.

Unilamellar vesicles with affinity ligands can be prepared either by coupling the ligand covalently to the reactive functional groups on the outer surface of the preformed liposomes or by first coupling it to a reactive phospholipid in an appropriate organic solution, which is then mixed with unmodified phospholipid in aqueous solution and sonicated. The ligand density on the liposome surface can be carefully controlled by simply varying the amount of derivatized phospholipid used during liposome preparation. Because of the small size of liposomes, all interactions of liposomes with proteins are kinetically controlled, and, moreover, since the ligands are on the outer surface of the liposome, there are no diffusional limitations during either adsorption or desorption step.

The feasibility of using liposomes as ligand carriers for membrane-based affinity separations was first studied with avidin as the model protein and biotin as the affinity ligand (Powers *et al.*, 1989a). Avidin was found to bind stoichiometrically to liposome-bound biotin in a 1:1 ratio. Based on the measured binding rate constants, the equilibrium binding constant was determined to be $6 \times 10^7 M^{-1}$, as compared to $10^{15} M^{-1}$ with the free biotin. The decrease in binding constant was attributed to the steric hindrance between the avidin molecule and the liposomes during binding as the biotin was immobilized directly on the liposome surface; however, the binding constant was sufficient for purification purposes.

Using the biotinylated liposomes, avidin was specifically isolated from model protein mixtures consisting of avidin and cytochrome *c*, and avidin and lysozyme, and also from its naturally occurring source, hen egg white (Powers *et al.*, 1989a,b). In the latter case,

hydrophobic impurities were first removed by centrifugation and ion-exchange chromatography, and subsequent affinity cross-flow filtration resulted in a recovery of 96% pure avidin with 80% of its original activity.

For the liposome-assisted affinity ultrafiltration process to be industrially feasible, the liposomes must be stable toward biological substances and have a relatively long shelf life. The susceptibility of liposomes to disruption by surfactants, hydrophobic molecules, and plasma proteins was overcome by the incorporation of large amounts of cholesterol (>33 mol %) into the lipid bilayer, leading to an enhanced stability and shelf life (Powers *et al.*, 1990).

The affinity liposomes with incorporated cholesterol were used for the purification of trypsin from a mixture of trypsin and chymotrypsin and from crude pancreatic extract (Powers *et al.*, 1990). The affinity ligand, *p*-aminobenzamidine (PAB), was coupled to the surface of the vesicles via a hydrophilic spacer arm, diglycolic acid, to prevent any steric hindrance during affinity binding. The equilibrium binding constant between trypsin and immobilized PAB, K_i , was shown to be dependent on the PAB density of the liposome surface. An increase in K_i was observed at high PAB loadings. Loss of trypsin during washing was thought to be due to the relatively low equilibrium constant of $1.7 \times 10^4 M^{-1}$ between trypsin and immobilized PAB. Elution of the bound trypsin was carried out with benzamidine as the eluent in a diafiltration mode to minimize the possibility of aggregation of liposomes. From the crude extract, approximately 68% of the starting trypsin was recovered with a purity greater than 98%. It was observed that the same PAB-liposome solution could be used without any loss in trypsin binding capacity for at least five successive trypsin purifications from trypsin/chymotrypsin mixtures over a 30-day period.

4.2. Water-Soluble Macromolecular Affinity Ligands

The advantages of water-soluble biospecific polymers as ligand carriers lie in the high yields obtained from ligand coupling, the rapid control of the amount immobilized, and the easy complexation with the complementary macromolecule, which implies a high binding capacity. Protein binding takes place under equilibrium conditions, resulting in quantitative complexation.

For the purposes of affinity ultrafiltration, ligands bound to water-soluble polymers necessitate the use of ultrafiltration membranes that have much smaller pores and hence much lower flux rates than the microfiltration membranes; accordingly, a longer processing time or a larger membrane surface area is needed. Both reversible and irreversible fouling can be a significant problem in filtration of polymeric solutions. On the other hand, soluble polymer is not likely to be degraded by attrition.

As shown in Table 1, high-molecular-weight dextran was used as ligand carrier in a number of studies on affinity filtration (Nylén, 1980; Hubert and Dellacherie, 1980; Adamski-Medda *et al.*, 1981; Choe *et al.*, 1986). A number of established coupling mechanisms can be employed for attaching affinity ligands to the carbohydrate polymer. Unfortunately, a serious problem faced during purification of trypsin from a mixture of trypsin and chymotrypsin using dextran-bound trypsin inhibitors was the nonspecific adsorption of the proteins to the polymer backbone (Adamski-Medda *et al.*, 1981; Choe *et al.*, 1986).

Luong and co-workers synthesized a high-molecular-weight (100,000) soluble polymer having an acrylamide backbone with pendant aminobenzamidine groups for the

purification of trypsin by affinity ultrafiltration (Luong *et al.*, 1988a; Male *et al.*, 1987). The polymer was found to be specific and effective in inhibiting trypsin.

The amount of inhibitor in the polymer could be varied by copolymerizing varying amounts of *N*-acryloyl-*m*-aminobenzamidine (referred to as the monomer) with acrylamide. The mechanism of trypsin inhibition by the affinity polymer was competitive, and the value of K_i appeared to depend on the acrylamide:monomer concentration ratio (Male *et al.*, 1987). At high monomer concentration, there was a decrease in the inhibition capacity of the polymer. An acrylamide:monomer ratio of 5:1 was found to be optimal for maintaining a balance between the number of available binding sites (high *M*-aminobenzamidine content) and the strength of the binding force (corresponding to low K_i).

The binding efficiency between the affinity polymer and trypsin was also dependent on the polymer concentration (Male *et al.*, 1987). Using an acrylamide:monomer ratio of 5:1, the amount of trypsin retained was about 90% at 0.75% polymer concentration. This level decreased to 54% when the concentration was reduced to 0.03%, even though the amount of enzyme introduced was also lowered by the same factor. This behavior was explained by the fact that decreasing the acrylamide concentration made the trypsin bound to the ligand more susceptible to the shear force resulting from the high flow rates through the membrane system. Though it may be possible to obtain a better binding efficiency using concentrated polymer solutions, that would most likely result in a very slow filtration rate due to the increased viscosity. Moreover, in the case of polymers exhibiting some nonspecific binding, these interactions may be modified in an unforeseeable manner with changes in concentrations of protein and macroligand.

The binding of the soluble affinity polymer with the enzyme was found to be instantaneous (Luong *et al.*, 1988b). This is an important observation as it suggests that the enzyme can be directly added to the polymer solution that is being passed through the ultrafiltration unit, instead of it being necessary to set up a holding tank for the binding to take place prior to ultrafiltration.

It was found important to maintain a neutral pH when the affinity polymer is used in conjunction with the ultrafiltration membrane plates. Both trypsin and chymotrypsin in a polymer solution passed through the ultrafilter very slowly at low and high pH values. Hence, acidic solution, which is often used to dissociate the trypsin-inhibitor complex during affinity chromatography, could not be used for elution from the soluble polymeric *M*-aminobenzamidine-enzyme complex.

The elution was found to be efficient when either arginine or benzamidine was used as the eluting agent. Trypsin with 98% purity and 90% yield was obtained in a batch process. The same polymer solution could be reused for more than one cycle of binding and elution. With the use of the same macroligand in a continuous process by recirculating the eluent as well as the ligand, trypsin was purified from crude porcine pancreatic extract with 97% purity and 77% yield. Based on the experimental data, a mathematical model has been developed to describe the dynamic behavior of the affinity cross-flow filtration process with this macromolecular ligand.

The polymer solution could be stored for several months without undergoing degradation or losing ligand molecules (Luong *et al.*, 1988c). The stored polymer exhibited the same binding capacity for trypsin as the freshly synthesized one. Also, the trypsin could be adsorbed by the polymer and stored until the enzyme was required for further processing.

The same soluble affinity polymer was used to purify urokinase from an artificial solution containing peroxidase and urokinase and from a crude urine source (Male *et al.*,

1990). In this case, the optimal acrylamide:monomer ratio was found to be 2:1. The process yields from the two sources were about 86% and 49%, respectively.

5. CONCLUDING REMARKS

In comparison to affinity chromatography, which is undoubtedly a superior analytical technique, affinity ultrafiltration is better suited for application on a large scale. Dilute feed streams can be handled in relatively short time periods. An affinity chromatography column is highly vulnerable to fouling and clogging, thereby reducing the system's productivity and limiting its use to fairly clear solutions. The ability of affinity ultrafiltration to process unclarified broths permits affinity interactions to be employed at the start of the downstream processing train such that the target protein is rapidly removed from a hostile environment.

The efficacy of the process is largely governed by the choice of the macromolecular affinity ligand—its selectivity, cost, and stability toward shear forces and recycling. It should be possible to extend the affinity ultrafiltration process to purify more than one biomolecule from a crude broth, either by using a group-specific ligand and manipulating the elution conditions or by employing different macroligands in a sequence.

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